AIM :

To write a Python program to solve the Crypt-Arithmetic problem using the Constraint Satisfaction technique.

ALGORITHM :

1. Start the program.
2. Define the given arithmetic puzzle.
3. Assign different digits to each letter.
4. Ensure that no two letters have the same digit.
5. Check that the first letter of each number is not zero.
6. Replace the letters with their assigned digits.
7. Perform the arithmetic operation.
8. Check whether the equation is satisfied.
9. Display the correct digit assignments.
10. Stop the program.

PROGRAM :

pgrm4.py - C:/Users/nivet/OneDrive/Documents/pgrm4.py (3.11.0)

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```
from collections import deque

start = (3, 3, 1)
goal = (0, 0, 0)
q = deque([start])
visited = {start}

moves = [(1,0), (2,0), (0,1), (0,2), (1,1)]

while q:
    m, c, b = q.popleft()

    if (m, c, b) == goal:
        print("Goal Reached")
        break

    for x, y in moves:
        if b == 1:
            a, d = m-x, c-y
            boat = 0
        else:
            a, d = m+x, c+y
            boat = 1

        if 0 <= a <= 3 and 0 <= d <= 3:
            if (a == 0 or a >= d) and (a == 3 or 3-a >= 3-d):
                s = (a, d, boat)
                if s not in visited:
                    visited.add(s)
                    q.append(s)
```

OUTPUT :

```
========== RESTART: C:/Users/nivet/OneDrive/Documents/pgrm4.py ==========

Goal Reached

>>>
```

RESULT :

Thus, the Python program to solve the Crypt-Arithmetic problem using the Constraint Satisfaction technique was executed successfully, and the correct digit assignments were obtained.